

# Worksheet 13

## Single phase induction motor - cap start

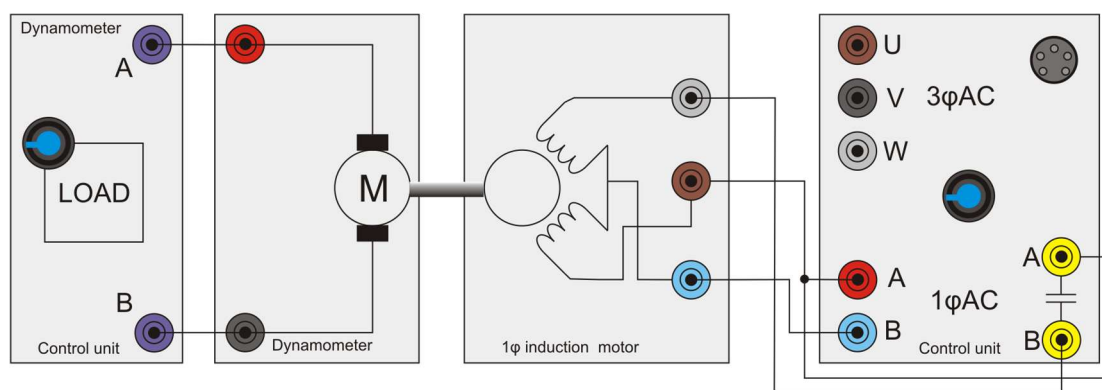
## Modern electrical machines



In many situations, engineers have a choice over which type of motors to use.

For simple AC applications, like pressure washers, they can use either an induction motor or a series motor.

The choice is governed by the electromechanical properties of the motor - the cost, size and long-term reliability required.



### Over to you:

- 1) Set up the single phase induction motor as shown in the diagram above.
- 2) On your PC load 'Open Log\_1phase.bat' and press 'Play'.
- 3) Use settings SINE, 20V, IMPROVED.
- 4) With the output frequency set at 50Hz, run the program and use the 'Sweep' function to take an automatic set of readings over a range of dynamometer loads.
- 4) When the sequence has finished, use Excel to plot a graph of speed vs torque.
- 8) Repeat this procedure for frequencies of 60, 70, 80, 90 and 100Hz.
- 9) On the next page, sketch graphs to show traces for all frequencies.

### So what?

How do you change direction of rotation?

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How do the speed / torque curves compare to the other motors you have looked at?

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What are the advantages and disadvantages of DC and AC machines?

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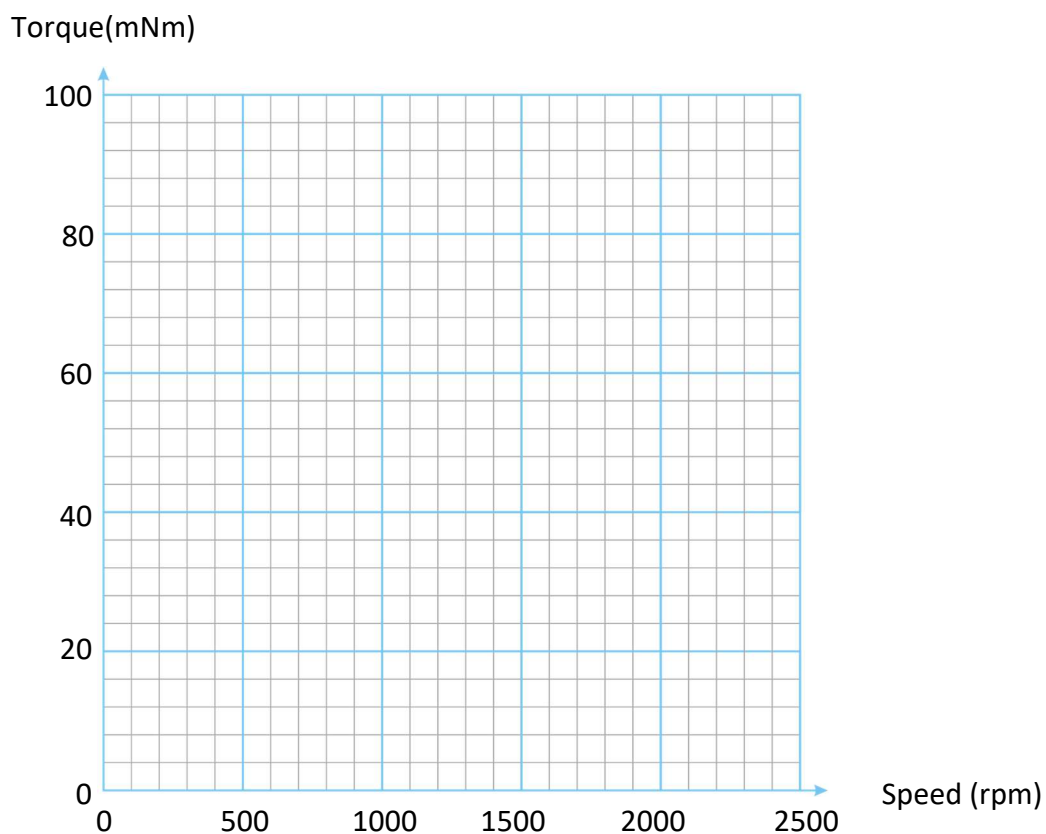


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The effect of speed on torque for different values of output frequency.